

pstat126_homework3

Melody Su

2026-05-28

```
install.packages("car", dependencies = TRUE)

## Installing package into '/home/aldo/R/x86_64-pc-linux-gnu-library/4.3'
## (as 'lib' is unspecified)

## also installing the dependencies 'curl', 'alr4', 'effects', 'rio'

## Warning in install.packages("car", dependencies = TRUE): installation of
## package 'curl' had non-zero exit status

## Warning in install.packages("car", dependencies = TRUE): installation of
## package 'rio' had non-zero exit status

data <- read.csv("Dataset for Homework 3-1.csv")
data$Exam_Level <- as.factor(data$Exam_Level)
data$Exam_Anxiety <- as.factor(data$Exam_Anxiety)

full_model <- lm(
  Exam_Score ~ Hours_Studied +
    Phone_Usage +
    Assignments_Completed +
    Exam_Level +
    Exam_Anxiety,
  data = data
)

summary(full_model)

##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Phone_Usage + Assignments_Completed +
##     Exam_Level + Exam_Anxiety, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.1827  -7.9433  -0.0638   6.9322  24.8682
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.60257     7.40373  -1.027  0.307152
## Hours_Studied     4.25804     0.62404   6.823 8.90e-10 ***
## Phone_Usage     -0.02523     0.54582  -0.046  0.963230
## Assignments_Completed  7.31612     0.72608  10.076 < 2e-16 ***
## Exam_LevelEasy  13.11194     2.93000   4.475 2.16e-05 ***
## Exam_LevelMedium 10.30498     2.69396   3.825 0.000236 ***
```

```
## Exam_AnxietyYes          1.52207      2.21913    0.686 0.494489
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.19 on 93 degrees of freedom
## Multiple R-squared:  0.7468, Adjusted R-squared:  0.7305
## F-statistic: 45.73 on 6 and 93 DF,  p-value: < 2.2e-16
```

1. Variable Selection

Use the AIC/BIC method to select the most appropriate predictors for the model

```
# AIC
aic_model <- step(full_model, direction = "backward")

## Start:  AIC=471.08
## Exam_Score ~ Hours_Studied + Phone_Usage + Assignments_Completed +
##      Exam_Level + Exam_Anxiety
##
##              Df Sum of Sq    RSS    AIC
## - Phone_Usage      1      0.2  9662.8 469.09
## - Exam_Anxiety      1     48.9  9711.5 469.59
## <none>                        9662.6 471.08
## - Exam_Level        2    2250.9 11913.5 488.03
## - Hours_Studied      1    4837.4 14500.0 509.67
## - Assignments_Completed 1   10548.9 20211.5 542.88
##
## Step:  AIC=469.09
## Exam_Score ~ Hours_Studied + Assignments_Completed + Exam_Level +
##      Exam_Anxiety
##
##              Df Sum of Sq    RSS    AIC
## - Exam_Anxiety      1      50.4  9713.3 467.61
## <none>                        9662.8 469.09
## - Exam_Level        2    2252.6 11915.4 486.04
## - Assignments_Completed 1   10592.9 20255.7 541.10
## - Hours_Studied      1   14345.2 24008.1 558.10
##
## Step:  AIC=467.61
## Exam_Score ~ Hours_Studied + Assignments_Completed + Exam_Level
##
##              Df Sum of Sq    RSS    AIC
## <none>                        9713.3 467.61
## - Exam_Level        2    2295.5 12008.8 484.82
## - Assignments_Completed 1   10572.4 20285.6 539.25
## - Hours_Studied      1   14346.3 24059.5 556.31

summary(aic_model)

##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Assignments_Completed +
##      Exam_Level, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -22.1330 -7.4058 0.3349 6.5607 25.9257
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.4503     3.6414  -2.046 0.043519 *
## Hours_Studied     4.2816     0.3615  11.845 < 2e-16 ***
## Assignments_Completed 7.3101     0.7189  10.169 < 2e-16 ***
## Exam_LevelEasy    13.1683     2.9050   4.533 1.69e-05 ***
## Exam_LevelMedium   10.4826     2.6517   3.953 0.000148 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.11 on 95 degrees of freedom
## Multiple R-squared:  0.7455, Adjusted R-squared:  0.7348
## F-statistic: 69.57 on 4 and 95 DF,  p-value: < 2.2e-16

AIC(aic_model)

## [1] 753.3955

# BIC
n <- nrow(data)
bic_model <- step(
  full_model,
  direction = "backward",
  k = log(n)
)

## Start:  AIC=489.32
## Exam_Score ~ Hours_Studied + Phone_Usage + Assignments_Completed +
##      Exam_Level + Exam_Anxiety
##
##              Df Sum of Sq    RSS    AIC
## - Phone_Usage      1      0.2 9662.8 484.72
## - Exam_Anxiety      1     48.9 9711.5 485.22
## <none>                    9662.6 489.32
## - Exam_Level        2    2250.9 11913.5 501.05
## - Hours_Studied      1    4837.4 14500.0 525.30
## - Assignments_Completed 1   10548.9 20211.5 558.51
##
## Step:  AIC=484.72
## Exam_Score ~ Hours_Studied + Assignments_Completed + Exam_Level +
##      Exam_Anxiety
##
##              Df Sum of Sq    RSS    AIC
## - Exam_Anxiety      1     50.4 9713.3 480.63
## <none>                    9662.8 484.72
## - Exam_Level        2    2252.6 11915.4 496.46
## - Assignments_Completed 1   10592.9 20255.7 554.13
## - Hours_Studied      1   14345.2 24008.1 571.12
##
## Step:  AIC=480.63
## Exam_Score ~ Hours_Studied + Assignments_Completed + Exam_Level
##
##              Df Sum of Sq    RSS    AIC
## <none>                    9713.3 480.63
```

```
## - Exam_Level          2      2295.5 12008.8 492.64
## - Assignments_Completed 1      10572.4 20285.6 549.67
## - Hours_Studied        1      14346.3 24059.5 566.73
```

```
summary(bic_model)
```

```
##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Assignments_Completed +
##      Exam_Level, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.1330  -7.4058   0.3349   6.5607  25.9257
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -7.4503     3.6414  -2.046  0.043519 *
## Hours_Studied     4.2816     0.3615  11.845 < 2e-16 ***
## Assignments_Completed  7.3101     0.7189  10.169 < 2e-16 ***
## Exam_LevelEasy    13.1683     2.9050   4.533 1.69e-05 ***
## Exam_LevelMedium   10.4826     2.6517   3.953 0.000148 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.11 on 95 degrees of freedom
## Multiple R-squared:  0.7455, Adjusted R-squared:  0.7348
## F-statistic: 69.57 on 4 and 95 DF,  p-value: < 2.2e-16
```

The best model according to both AIC and BIC was:

Exam Score ~ Hours Studied + Assignments Completed + Exam Level

The variables **Phone Usage** and **Exam Anxiety** were removed because they did not improve the model sufficiently.

```
final_model <- lm(
  Exam_Score ~ Hours_Studied +
    Assignments_Completed +
    Exam_Level,
  data = data
)
AIC(final_model)
```

```
## [1] 753.3955
```

```
BIC(final_model)
```

```
## [1] 769.0266
```

Best model statistics:

- AIC = 753.3955
- BIC = 769.0266

2. Multicollinearity Check

Apply the Variance Inflation Factor (VIF) technique to detect and remove multicollinearity among the predictors.

```
library(car)
```

```
## Loading required package: carData
```

```
vif(aic_model)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## Hours_Studied    1.042960  1      1.021254
## Assignments_Completed 1.011854  1      1.005909
## Exam_Level       1.052601  2      1.012899
```

The VIF values for **Hours Studied**, **Assignments Completed**, and **Exam Level** are all below 5, indicating that multicollinearity is not a concern. Therefore, no variables need to be removed based on the VIF analysis.

3. Final Model Specification

Write down the final reduced model obtained after applying these techniques.

$$\widehat{\text{Exam Score}} = -7.4503 + 13.1683(\text{Easy}) + 10.4826(\text{Medium}) + 4.2816(\text{Hours Studied}) + 7.3101(\text{Assignments Completed})$$

4. Interpretation of Estimates

Provide a statistical interpretation of the coefficients output (estimates, standard error, t-value) in the final model.

Hours Studied: For every additional hour studied, the expected exam score increases by about 4.28 points, holding all other variables constant

The standard error is small (0.361), meaning the estimate is precise

The large t-value (11.845) and very small p-value indicate that **Hours Studied** is highly significant.

Assignments Completed: For every additional assignment completed, the expected exam score increases by about 7.31 points, holding other variables constant.

The large t-value (10.169) shows strong statistical significance.

Exam Level: Compared to a difficult exam:

- students taking an easy exam score about 13.17 points higher on average
- students taking a medium exam score about 10.48 points higher on average

Both coefficients are statistically significant

Intercept: The intercept represents the predicted exam score when all predictors equal zero and the exam is difficult.

Although the intercept is statistically significant, it may not have a strong practical interpretation because some predictor values may not realistically equal zero.

5. Hypothesis Testing

(a) Partial Significance Tests

Explain the significance of individual coefficients based on their p-values. (This test assesses the significance of individual predictors in the model by checking whether each coefficient is significantly different from zero).

For each predictor:

$$H_0 : \beta_j = 0 \quad H_A : \beta_j \neq 0$$

Hours Studied: significant

Assignments Completed: significant

Exam Level (Easy): significant

Exam Level (Medium): significant

Since all p-values are below 0.05, all predictors in the final model are statistically significant.

(b) Global Significance Test

Interpret the F-statistic to assess the overall significance of the model, (This test evaluates whether at least one predictor in the model is significant).

$$F = 69.57$$

with p-value:

$$2.14 \times 10^{-27}$$

Hypothesis:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \quad H_A : \text{At least one } \beta_j \neq 0$$

Since the p-value is extremely small, we reject the null hypothesis.

Conclusion: The regression model is globally significant, meaning at least one predictor is useful for explaining exam scores.

6. Model Performance

(a) interpret R^2 and Adjusted R^2 to assess model fit

Model statistics:

- $R^2 = 0.746$
- Adjusted $R^2 = 0.735$

Interpretation:

- The model explains approximately 74.6% of the variability in exam scores.
- The adjusted R^2 is slightly smaller because it accounts for the number of predictors used.
- Since both values are relatively high, the model provides a good fit to the data.

(b) Residual Standard Error (RSE)

The Residual Standard Error is:

$$RSE \approx 7.37$$

Interpretation:

- The RSE measures the typical prediction error of the model.
- On average, the predicted exam scores differ from the actual scores by about 7.37 points.
- A smaller RSE indicates better prediction accuracy.

Overall, the model performs well and explains a substantial portion of the variation in exam scores.